Final Year Project (Year 22)
Feasibility Study

Electronic and Telecommunication Engineering

A MULTI-RF HEAD RECEIVER ARRAY FOR SENSING AND COMMUNICATIONS

Supervisors :

Prof. Arjuna Madanayake

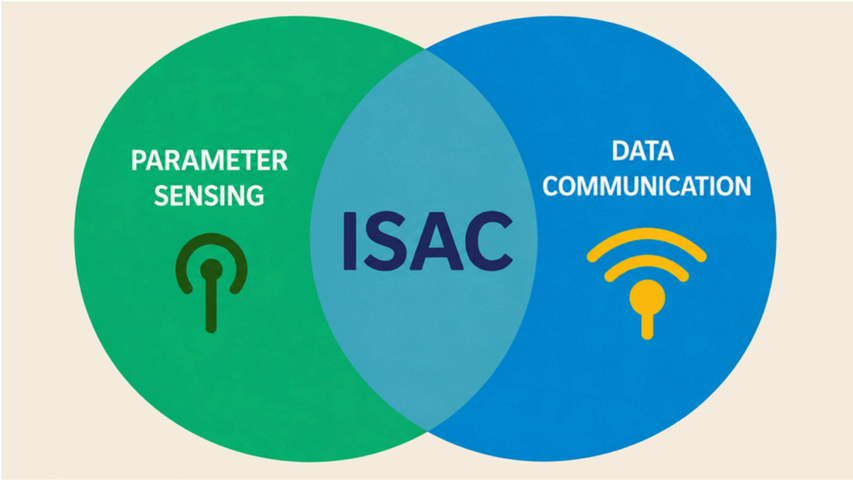
Dr. Chamira U. S. Edussooriya

Group No.20 Members :

Ranaweera R.P.D.D.H.	220511N
Weragoda W.A.A.P.	220692R
Anuradha D.M.B.P.	220036L
Peiris P.I.U.	220452H

Motivation

Future promising **6G** communication model expects seamless **connectivity** and **awareness of its surroundings**.



Integrated Sensing And Communication (ISAC) can enable simultaneous communication and sensing functionality, promising seamless connectivity.

Motivation - What is ISAC ?

ISAC - Integrated Sensing and Communication,

- employs a **shared hardware platform**
- utilize **shared waveform**
- **unified signal processing** chain

to perform simultaneous sensing and wireless data communication.

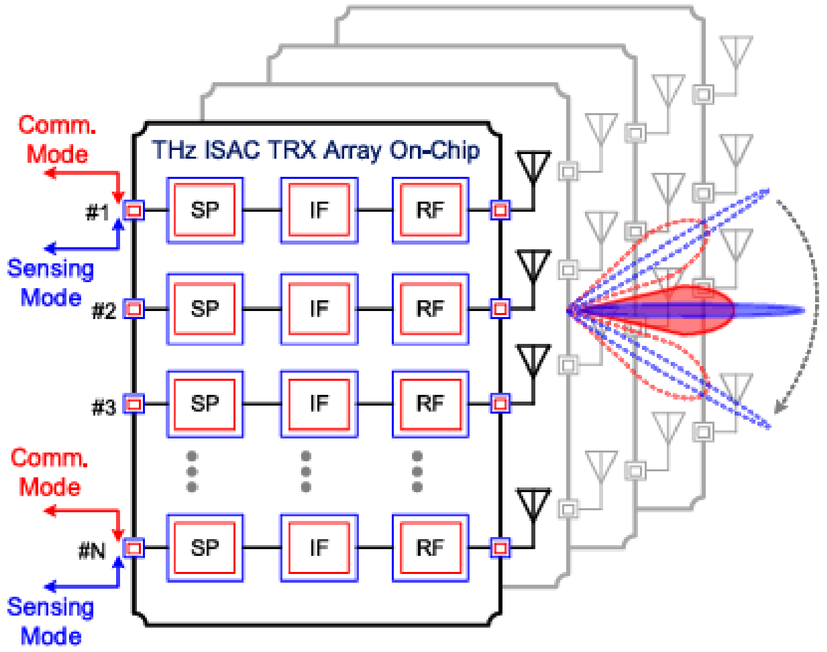
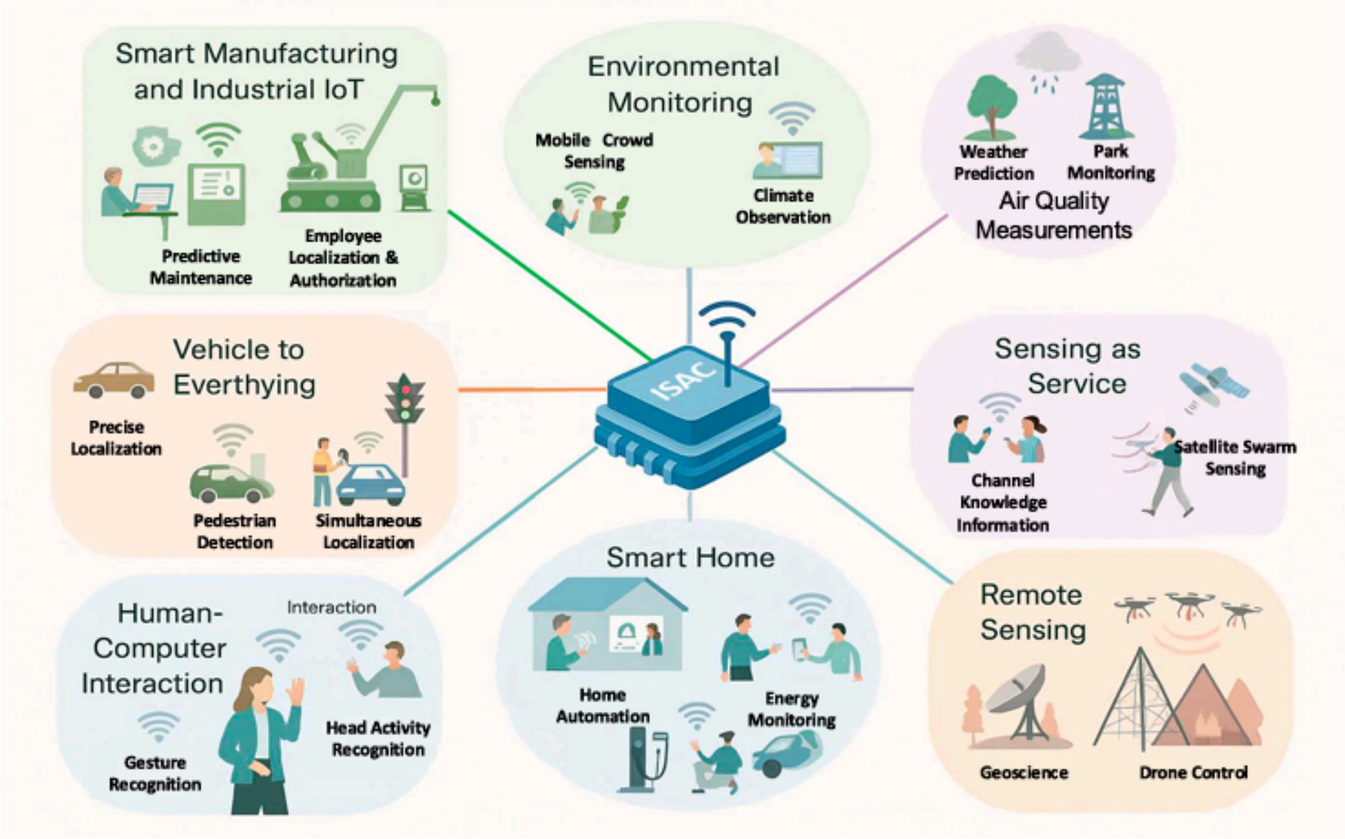


Illustration of ISAC concept

K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025, doi: 10.1109/JSTEAP.2025.3610564

Motivation - What is ISAC ?



K. Wu, Y. Bigdeli, S. A. Keivaan, J. Deng and P. Burasa, "Integrated Sensing and Communication (ISAC) Transceiver: Hardware Architectures, Enabling Technologies, and Emerging Trends," in IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, vol. 1, no. 1, pp. 37-64, Sept. 2025, doi: 10.1109/JSTEAP.2025.3610564

Problem Statement

Design a multi-head RF receiver capable of ISAC

Why do we need such reciever ?

- Need for simultaneous communication and sensing.
- Optimized multi-functional capability,
  - e.g., cellular communication, location intelligence, spectrum sensing etc.
- Reduced hardware duplication.

Primary Objectives

Building a **multi-head RF chain** and **three antenna arrays** for both sensing and communication without much hardware redundancy.

Implement **simultaneous communication and UAV detection**.

Scope

Multi-head RF receiver

- Simulate and design of **antenna arrays (two Vivaldi arrays and one patch antenna array)**.
- Simulate and design of RF chains for communication and sensing.
- Perform testing and verify the functionality of the receiver in the field.

Communication and sensing implementation

- Verify sensing capabilities using Orthogonal Time Frequency Space modulation (OTFS).
- Enable multi-band communication.

Scope - what is OTFS ?

The problem with standard signals,

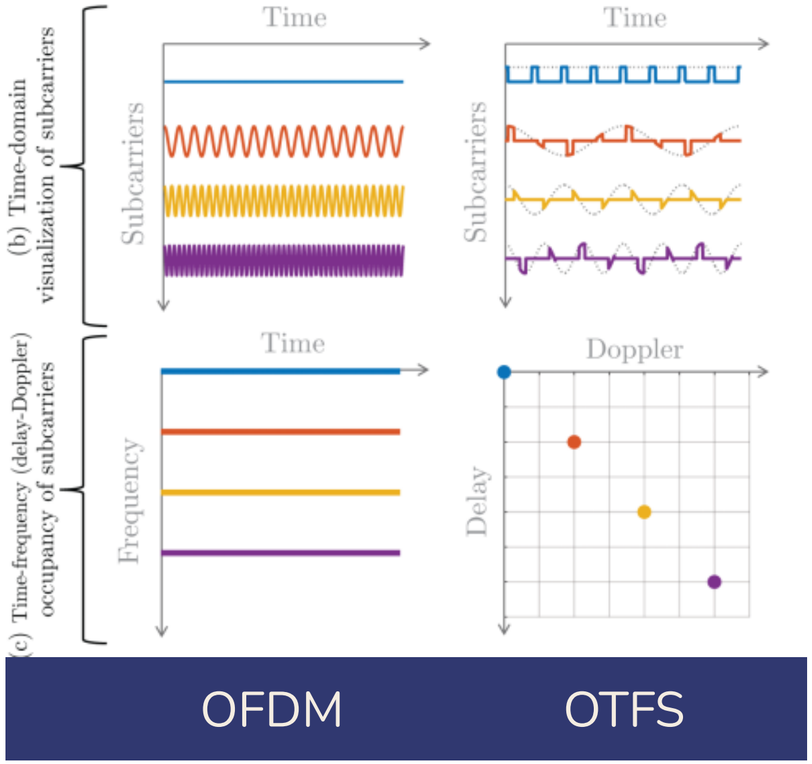
State-of-the-art Orthogonal Frequency Division Multiplexing (OFDM) is **unstable to severe Doppler shifts** which cause Inter-Carrier Interference (ICI).

- e.g., fast moving targets like drones creates sever Doppler shifts

OTFS solves this by mapping the data into the Delay-Doppler domain.

How it works?

It modulates information onto a localized pulse in **Delay-Doppler domain**, which **immunes** for larger Doppler shifts.



represntation of OFDM and OTFS on time and time frequency domain

Therefore OTFS become prominent in future highly dynamic communication topologies.

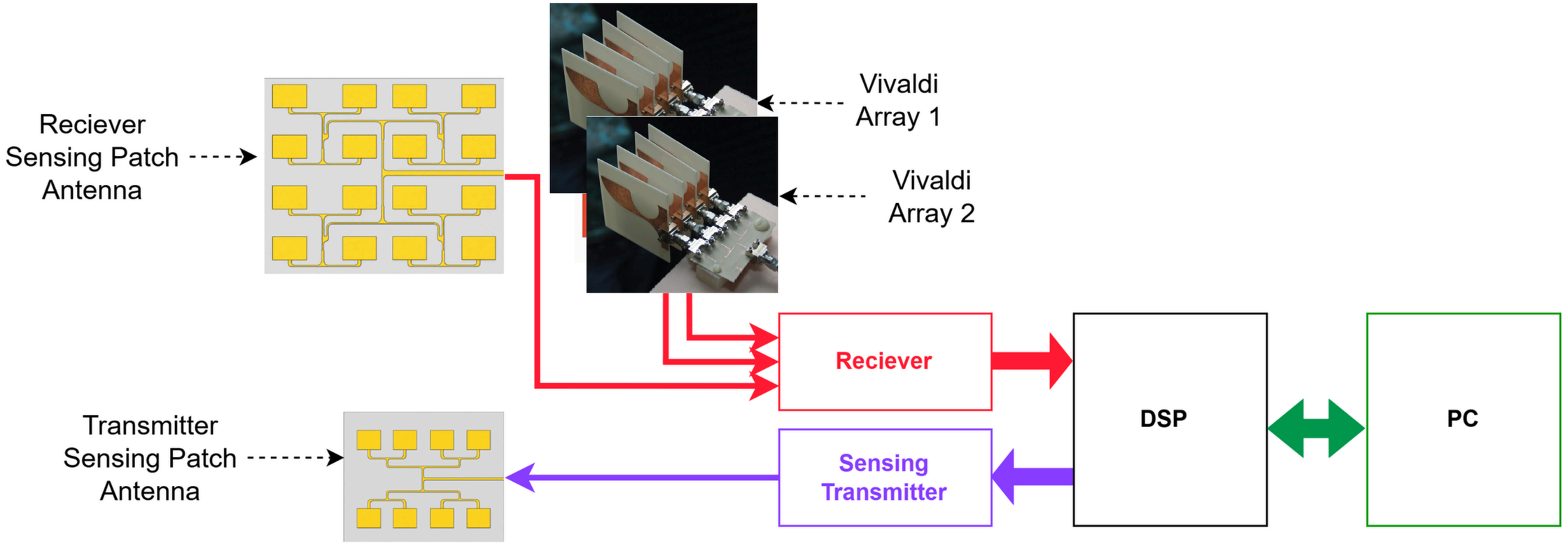
Delay-Doppler domain - A coordinate system that maps channel paths depending on their path delay and Doppler shifts

Rou, Hyeon Seok & Ranasinghe, Kuranage Roche Rayan & Savaux, Vincent & Abreu, Giuseppe & G., David & Masouros, Christos. (2026). Affine Frequency Division Multiplexing (AFDM) for 6G: Properties, Features, and Challenges. IEEE Communications Standards Magazine. 10. 216-225. 10.1109/MCOMSTD.2025.3643183.

Literature review

Paper	Hardware implementation	Frequency / Waveform	Communication + Sensing arrangement	Limitation / Gap
Pegoraro *et al.*, "HiSAC: High-Resolution Sensing with Multiband Communication Signals"	RFSoC + 16-element Sivers phased arrays	58–64 GHz, multiband communication signals	Communication pilots are reused for sensing	Bands are all within the same mmWave region
Wang *et al.*, "An RFSoC-Based Scalable OTFS Prototyping Platform for Integrated Sensing and Communications"	RFSoC-based physical prototype	OTFS - Orthogonal Time Frequency Space	Same OTFS platform performs communication and sensing	Main flexibility is waveform/baseband
George *et al.*, "Hardware-Software Platform Enabling Joint Communication and Radar Sensing at 25 GHz with 1 GHz Bandwidth"	Multi-purpose RX front-end, antennas and hardware/software platform	5G NR n258, ~25 GHz, 1 GHz Bandwidth	Receiver front-end, antenna and waveform are configured for communication and radar sensing	One operating RF band
Wei *et al.*, "UAV's Rotor Micro-Doppler Feature Extraction Using Integrated Sensing and Communication Signal: Algorithm Design and Testbed Evaluation"	Hardware ISAC base-station testbed + real DJI M300 RTK UAV	Sub-6-GHz 5G/ISAC	ISAC signal is used for communication and extracting UAV rotor micro-Doppler	Application-specific fixed ISAC hardware
da Gomez *et al.*, "Experimental Testbed for Frequency-Division Multiplexed Joint Radar–Communication System Validation in D-Band Using a 65-nm CMOS Communication Transceiver"	65-nm CMOS wideband transceiver and experimental JRC testbed	116–170 GHz D-band, Frequency Division Multiplexing	Radar and communication use different frequency resources within the same hardware platform	Both functions remain inside one D-band range

Project architecture



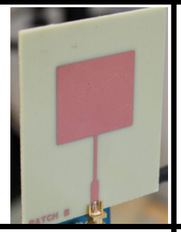
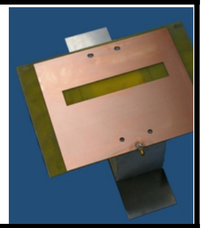
Current Project Architecture

Antenna arrays

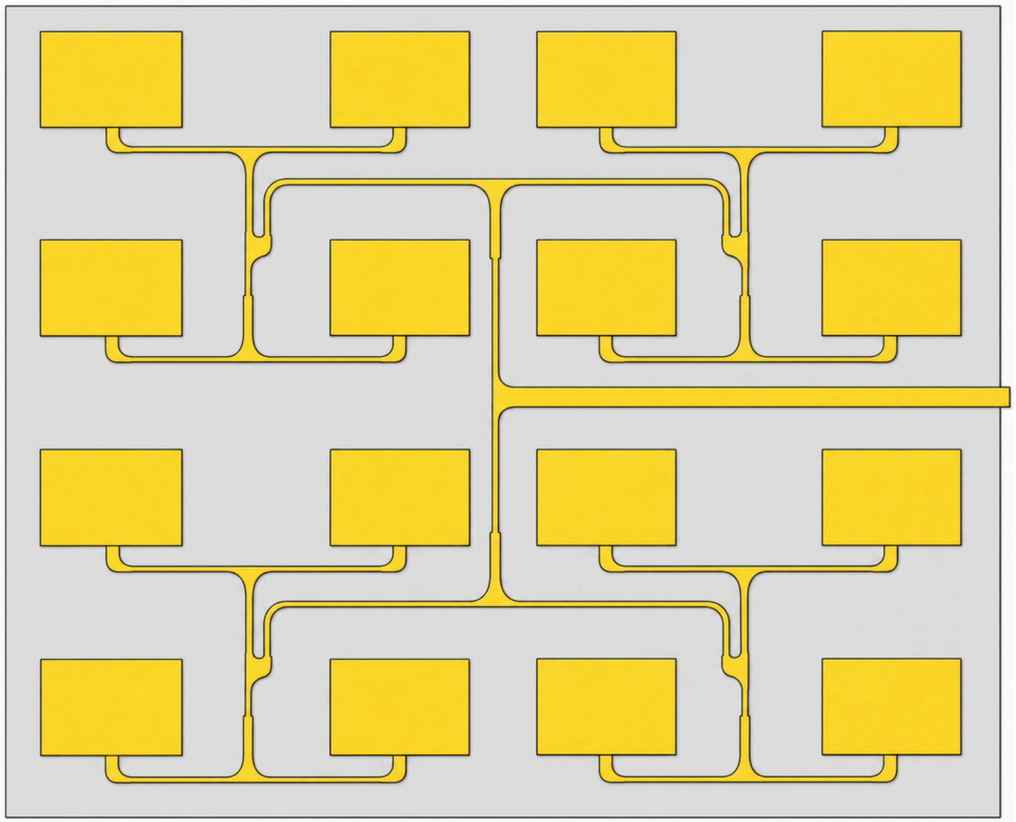
Antenna array	Operating frequency range	Bandwidth	Center frequency	Suitable applications
Vivaldi array 1	**800–1600 MHz**	**800 MHz**	**1.2 GHz**	**800/900 MHz cellular bands**, L-band communication/telemetry, **GNSS**
Vivaldi array 2	**1000–3000 MHz**	**2 GHz**	**2.0 GHz**	**Cellular communication** around 1.8/2.1/2.6 GHz, **2.4 GHz Wi-Fi/Bluetooth**
Patch antenna array	**5.7–6.0 GHz**	**300 MHz**	**5.85 GHz**	**Doppler** and **micro-Doppler extraction** for drone identification

Patch antenna design

Why do need to use a patch antenna ?

Design criterion	Rectangular patch	Yagi-Uda	Slot antenna
Basic structure	Conductive patch printed above a ground plane 	Driven element with reflector and directors 	Slot cut into a conductive ground plane 
Bandwidth	Naturally narrowband	Narrow to moderate bandwidth	Narrowband or broadband
Physical profile	Low and compact	Longer and protruding	Low and planar
PCB integration/ Fabrication	Excellent/ simple	Normally requires separate mechanical structure	Excellent/ PCB-compatible
Main advantage	Compact, planar, and easy to integrate	High directional gian	Low profile with flexible geometry

Patch antenna design



Proposed patch antenna array design

Why do we use an antenna array of 16 element patches ?

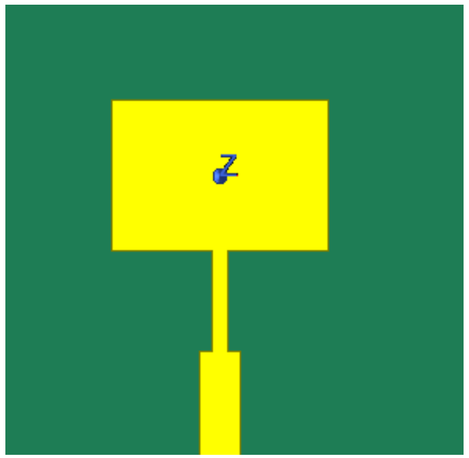
- The proposed design is a ***16-element (4 × 4) corporate edge-fed*** microstrip patch antenna array, impedance-matched using ***quarter-wave transformers***.
- An antenna array provides a **higher directivity** and **gain, narrower beamwidth**, **controllerable beam direction** and **control over sidelobes and nulls** etc.
- Ideally, ***D_"array" ≈ N * D_"element"*** **.**
- A 4 x 4 array provides benefits such as ***two dimensional beam control***, ***two-axis steering***, ***balanced E/H beamwidths***, ***compact aspect ratio*** etc.

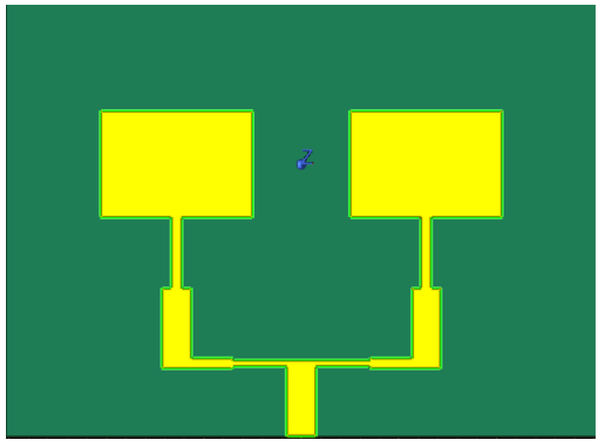
Patch antenna design

Evolution of patch antenna array design and simulation

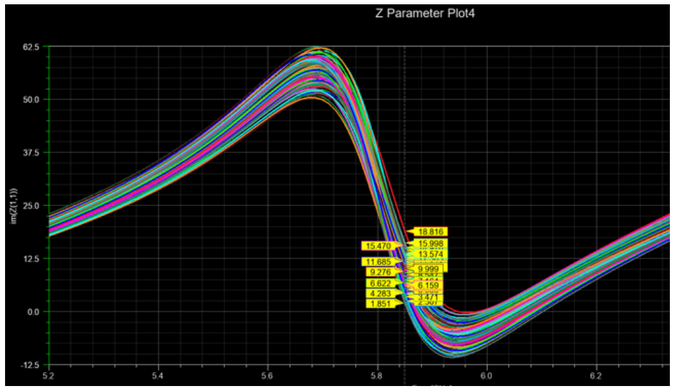
	Name	Value	Unit	Evaluated Value
	f0	5.85	GHz	5.85GHz
	er	4.4		4.4
	Wp	1000 * c0 / (2 * f0) * sqrt(2 / (er + 1))	mm	15.593836mm
	h	1.6	mm	1.6mm
	e_eff	(er + 1)/2 + ((er - 1)/2) * (1 + 12*h/Wp)^(-0.5)		3.8380838
	dL	0.412 * h * ((e_eff + 0.3) * (Wp/h + 0.264)) / ((e_eff - 0.258) * (Wp/h + 0.8))		0.72321916mm
	Lp	((c0)/(2*f0*sqrt(e_eff)) - 2*dL - 0.764654mm)*1000	mm	10.868mm
	lambda0	1000 * c0 / f0	mm	51.246574mm
	Sx	lambda0 / 2		25.623287mm
	Sy	Sx		25.623287mm
	MarginX	15	mm	15mm
	MarginY	15	mm	15mm
	BoardX	3 * Sx + Wp + 2 * MarginX		122.4637mm
	BoardY	3 * Sy + Lp + 2 * MarginY		117.73786mm
	Wtest	0.5	mm	0.5mm
	c	c0		2.9979246e+08
	Lp_sweep	10.94	mm	10.94mm
	Wp_sweep	15.594	mm	15.594mm
	WM	1.1	mm	1.1mm
	LM	7.25	mm	7.25mm
	W50	3	mm	3mm

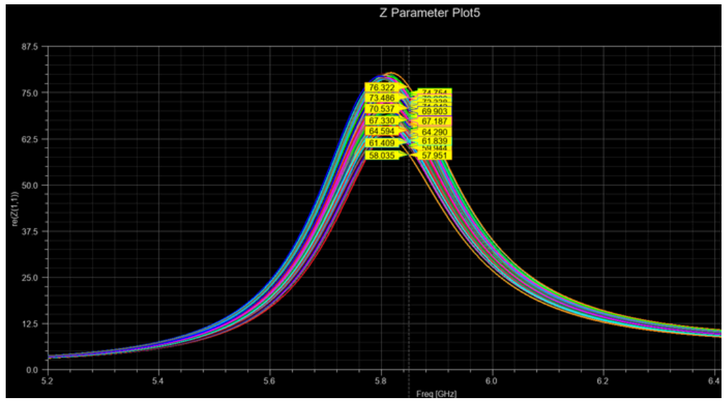
Design equations and parameter values



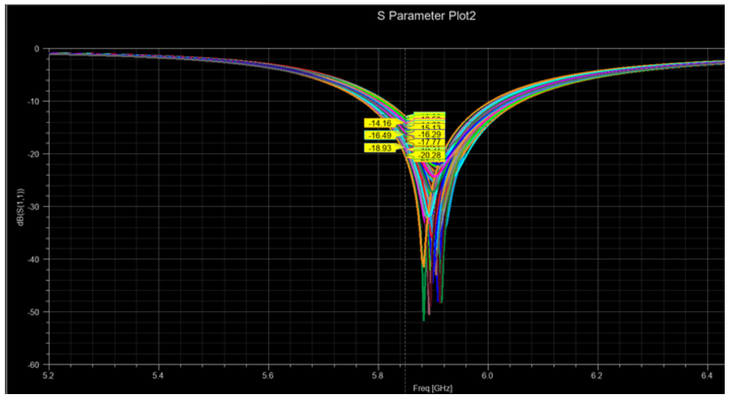


Tuned single and dual patch elements designs





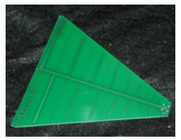
Antenna edge impedance plots re(Z 1,1) + j im (Z 1,1)



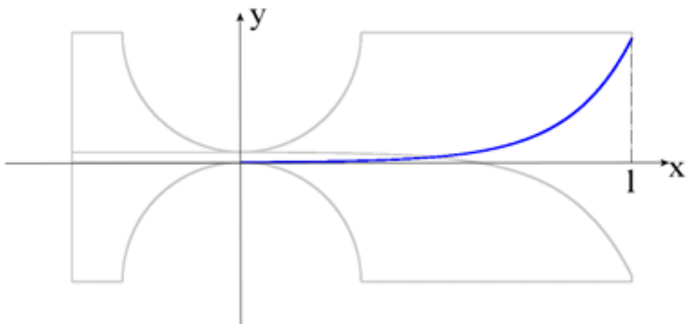
Antenna S 1,1 parameter plot

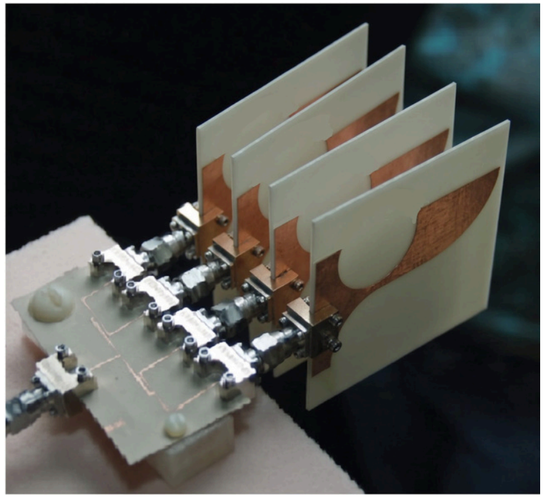
Antipodal Vivaldi antenna design

Why do need to use a Vivaldi antenna?

Design criterion	Horn antenna	Log Periodic antenna	Tapered-Slot Vivaldi	Antipodal Vivaldi
Basic structure	Flared waveguide 	Multiple driven and parasitic elements 	Exponentially tapered slot 	Overlapping antipodal conductors with tapered slot 
Bandwidth	Very wide	Very wide	Ultra-wideband	Ultra-wideband
Physical profile	Bulky and large	Long and protruding	Low-profile and planar	Low-profile and planar
PCB integration/ fabrication	Difficult	Moderate	PCB-compatible	PCB-compatible
Main advantage	High gain and directivity	Very wide frequency coverage	Wideband and low profile	Wideband, compact

Antipodal Vivaldi antenna design





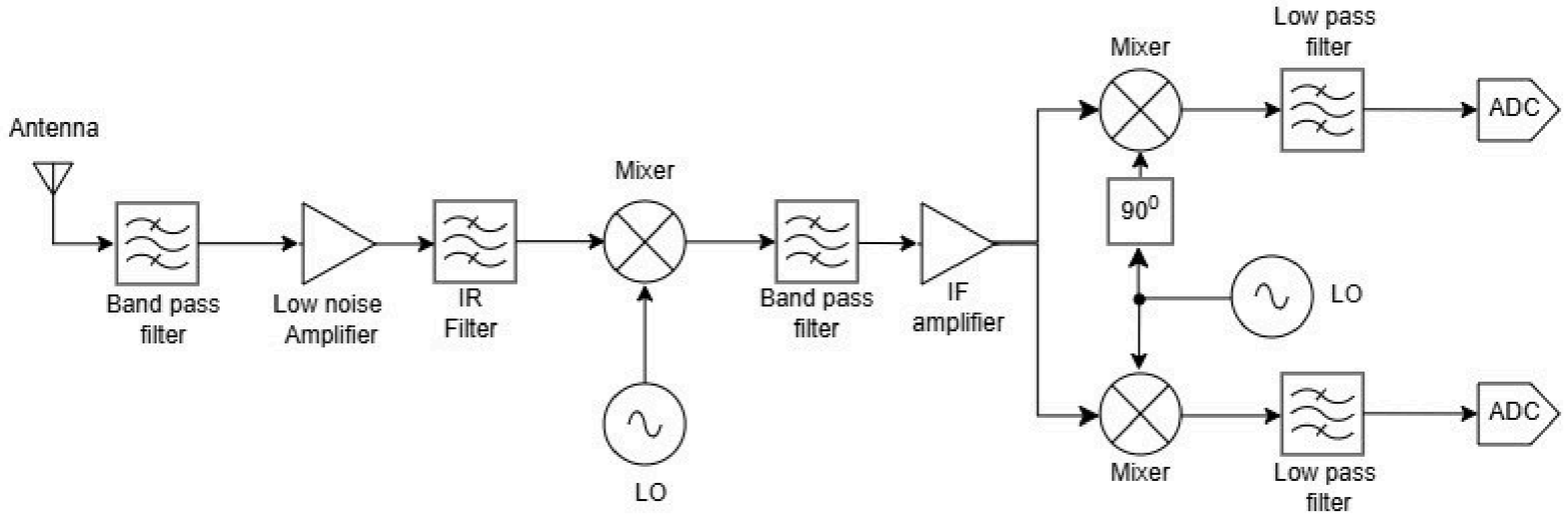
Why do we use an antipodal Vivaldi antenna?

- Wide-band operation with broad impedance and gain bandwidth.
- Directional end-fire radiation with good gain and directivity.
- Smooth feed-to-radiator transition for improved impedance matching.

Antenna array: higher effective aperture and directivity for improved communication performance.

G. Vinci and R. Weigel, "Multiband Planar Vivaldi Antenna for Mobile Communication and Industrial Applications," 2010 International Conference on Electromagnetics in Advanced Applications, Sydney, NSW, Australia, 2010, pp. 93-96

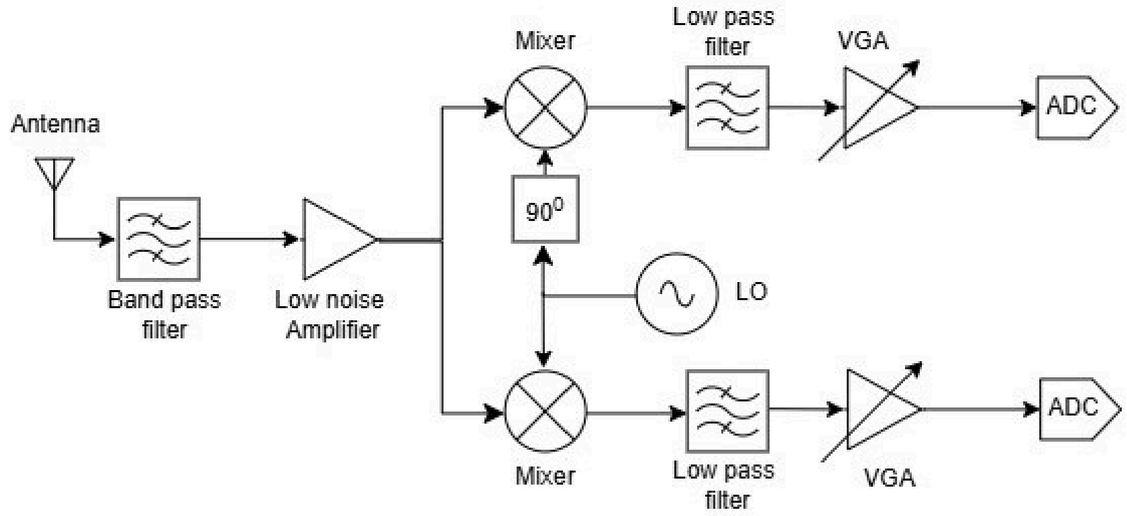
Hetrodyne architecture



- High sensitivity and selectivity through RF/IF filtering and amplification.
- Multiple frequency conversions provide flexible gain distribution and good interference rejection. Image frequencies reappear because the IF is non-zero.
- Need to apply image-rejection filters

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

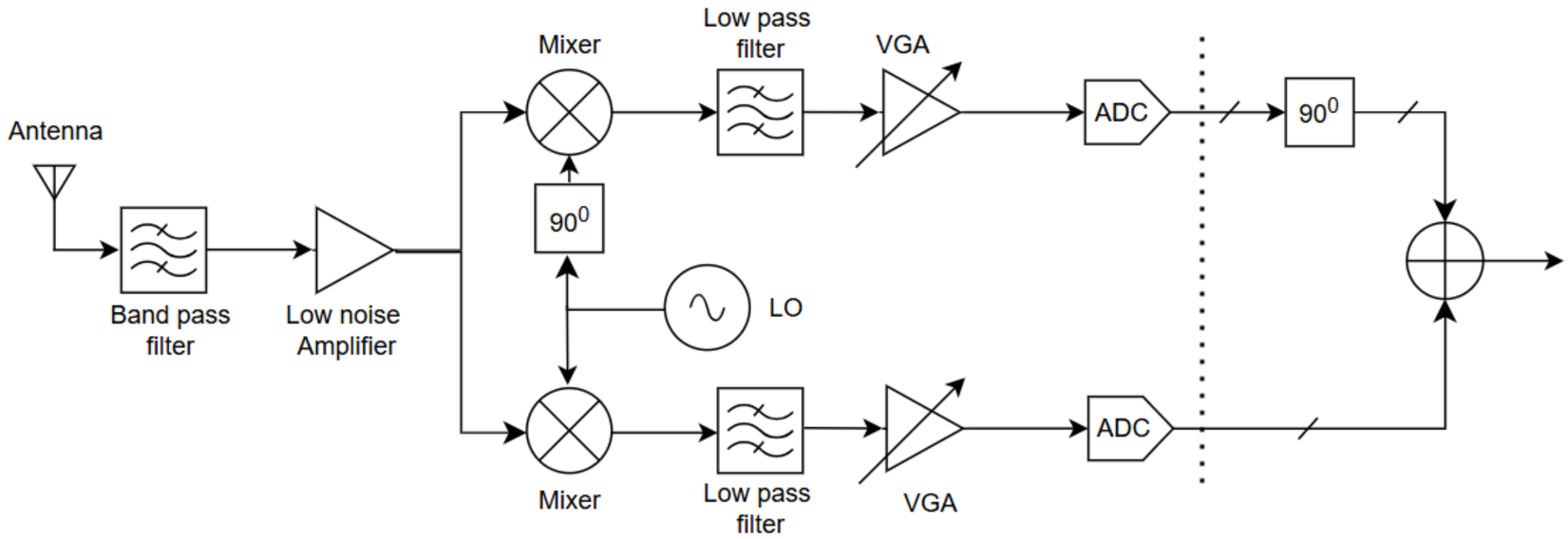
Direct conversion architecture



- Directly converts RF to baseband.
- Simple, low-power, and highly integrated architecture suitable for multi-standard applications.
- DC offset, flicker noise, LO leakage, and self-mixing can degrade the received signal.

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

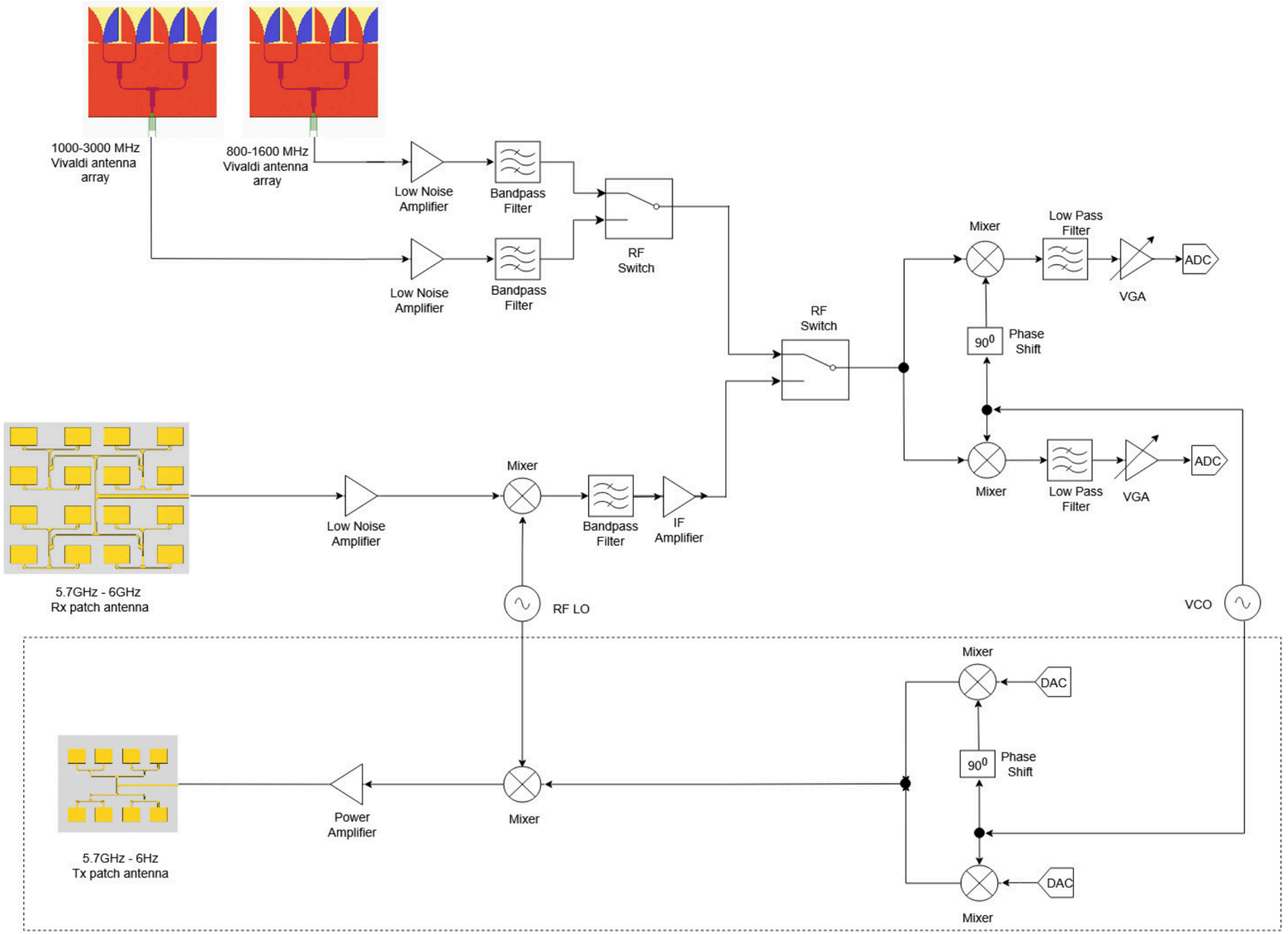
Low IF architecture



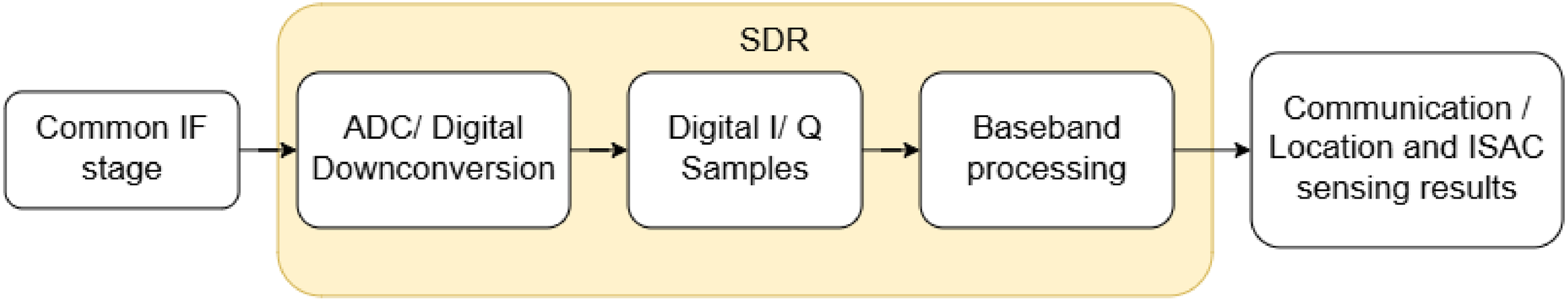
- Converts RF to a low intermediate frequency.
- Image frequencies reappear because the IF is non-zero.
- Image-rejection filtering or mixers are required, increasing implementation complexity.

B. Razavi, “RF Microelectronics,” 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2012.

Receiver Architecture (RF chain)



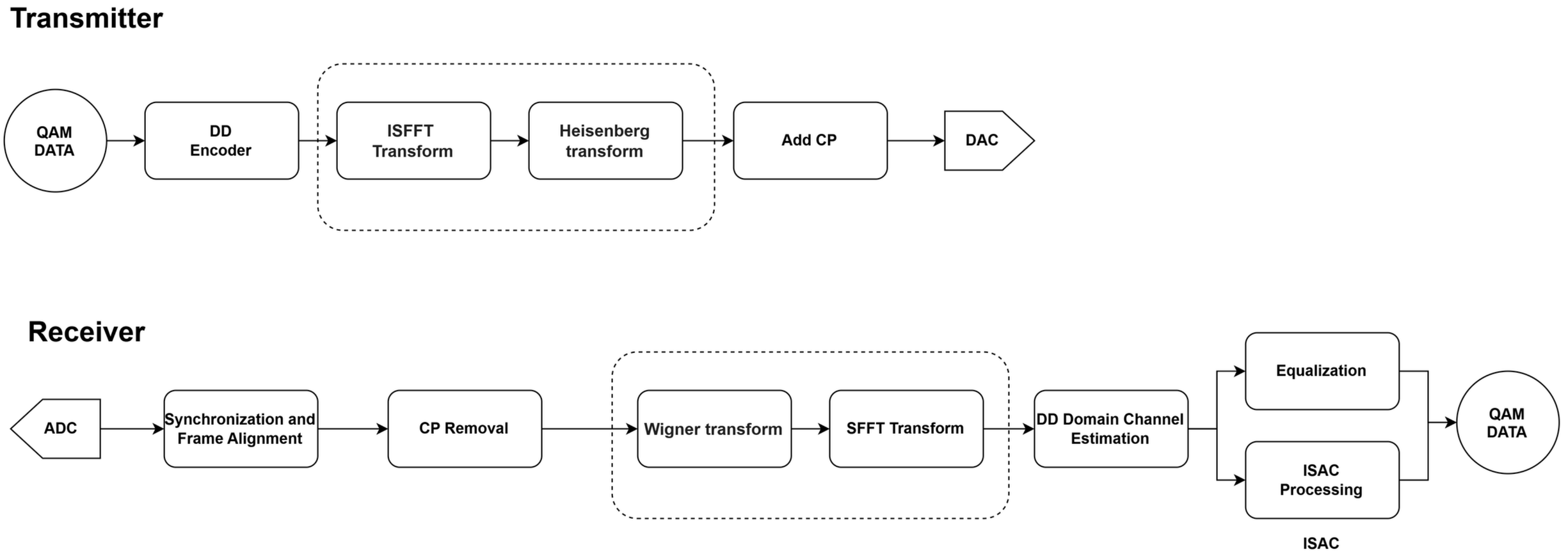
DSP



Three baseband processing paths

- 800 - 1600MHz : Processes GSM/GMSK signals for location intelligence.
- 1000 - 3000MHz : Processes cellular signals.
- OTFS processor : Processes ISFFT- OTFS signals in the 5.7GHz - 6.0GHz band.

ISFFT-OTFS baseband processing block diagram



Xiang, J., Hao, S. & Qi, L. A BEM-OTFS channel estimation method for high mobility 6G-V2X. SIViP 18, 6799–6806 (2024). https://doi.org/10.1007/s11760-024-03352-5
https://in.mathworks.com/help/comm/ug/otfs-modulation.html

Uniqueness

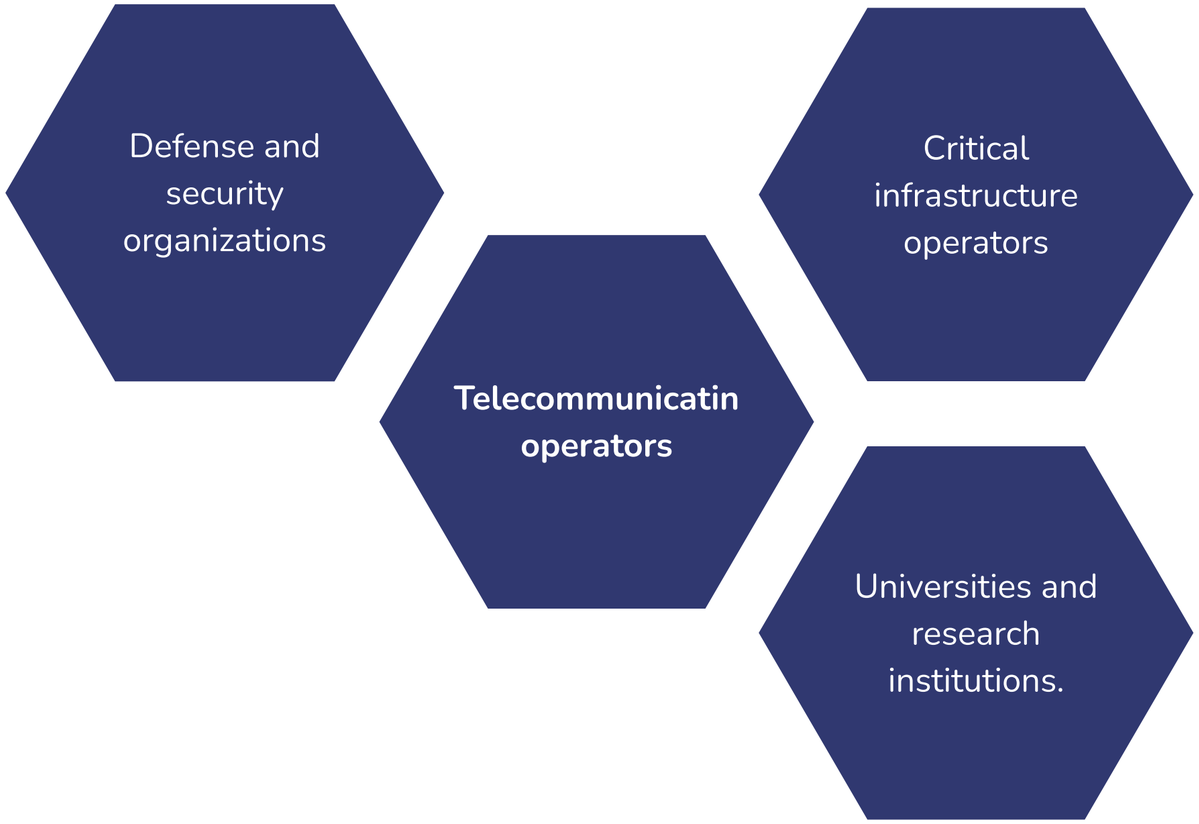
Multifunctional receiver platform

1. Can accommodate multiple communication bands

- e.g.,
    - 800MHz - 1600MHz band - GSM communication
    - 1000MHz - 3000MHz band - Wi-Fi and Bluetooth communication

2. OTFS modulation is used for sensing application

Beneficiaries



Risk factors

- Mutual coupling & RF interference
  - Risk: nearby antennas interfere with each other.
  - Mitigation: optimize spacing and isolation.
- Time constraints
  - Risk: delay in component procurement, manufacturing, or model convergence.
  - Mitigation: parallel task execution and phased integration to reduce bottlenecks.
- Legal / regulatory feasibility
  - Risk: transmitter power may surpass the power limitations imposed by the TRCSL.
  - Mitigation: carefully design the system not to violate power regulations.
- Hardware failure
  - Risk: PCB fabrication defects, antenna mismatches, or component damage.
  - Mitigation: simulate all designs thoroughly and fabricate from certified company .

Bill of Materials

Component	Part	Qty	Unit Price (USD)	Cost for Qty
Mixer	**HMC219BMS8GE**	3	**$18.14**	**$54.42**
Tx I/Q Modulator	**ADL5375-05ACPZ-R7**	1	**$15.81**	**$15.81**
Low Noise Amplifier	**HMC717ALP3E**	1	**$17.87**	**$17.87**
Receiver Mixer	**HMC219BMS8GE**	1	**$18.14**	**$18.14**
IF Amplifier	**ADL5611ARKZ-R7**	1	**$6.45**	**$6.45**
Vivaldi-head LNA	**ADL5523ACPZ-R7**	2	**$4.96**	**$9.92**
RF Head Selector	**SKY13373-460LF**	1	**$1.47**	**$1.47**
Common I/Q Receiver	**LTC5594IUH#PBF**	1	**$44.94**	**$44.94**
Fixed 4 GHz LO	**CVCO55CC-4000-4000**	1	**$41.89**	**$41.89**
Tunable 800–3000 MHz LO	**ADF4351BCPZ**	1	**$23.05**	**$23.05**
Passive Components				**$35.00**
PCB , Antenna Array				**$50.00**
Total				**$268.96**

Task delegation

Tasks	Agrajith	Bathiya	Dulina	Ishan
Antena design and simulation				
ISAC algorithm design				
SDR implementation				
Receiver design				
PCB Design				
General				
Documentation				

Project deliverables

- **Multi-RF head receiver array** with a 4-channel PCB, antennas, and RF front ends.
- **Embedded software and processing** for receiver control, configuration, and received-data processing.
- **Conference or journal paper** submitted to a recognized publication venue.
- **Final project documentation** including the report, presentation, demonstration video, and technical documents.

Resource Requirements

Software tools

- RF simulation tools (ADS, HFSS, or equivalent), Altium PCB designer.
- Xilinx Vivado/ Vitis HLS.
- MATLAB/Python-based signal processing.

MATLAB®

Ansys / HFSS

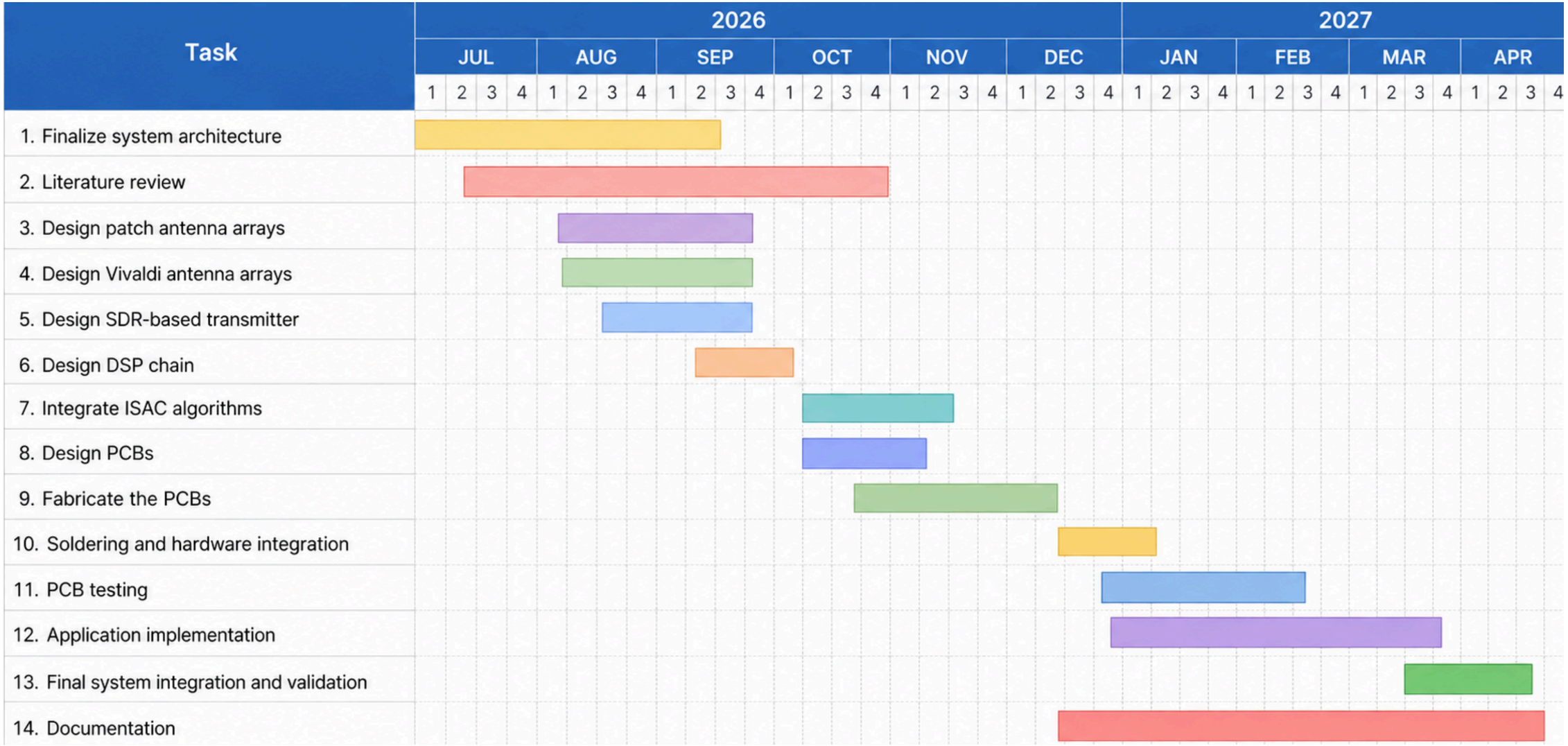
Altium Designer.

Hardware tools

- RF LNAs, mixers, switches, tunable filters, and IF amplifiers
- Spectrum analyzer, VNA, signal generator, and RF measurement equipment
- FPGA , SDR for digital signal processing



Project timeline



MORE INFO

Why multiple antenna sub arrays

- In antenna array the spacing between two antennas decided by the upper bound frequecy wave length of the desired band. (Sampling the wave phases without aliasing)
- Antenna array elements are equally spaced (linear array)

800MHz - 6000MHz band
d - space between two elements

$$d = \lambda/2$$

lambda - free space wave length of the upper bound frequency of the frequency band

$$\lambda_{6G} = \frac{c}{6 \times 10^9} = 50 \text{ mm}$$

$$\lambda_{800M} = \frac{c}{800 \times 10^6} = 375 \text{ mm}$$

- Therefore the spacial resolution for the lower frequencies is lower. For lower frequencies need a larger electrical aperture. Need a larger number of antenna elements.
- It is too costly

Solution :- Divide the the frequency range into proper sub ranges and allocate differnt antenna arrays.

https://en.wikipedia.org/wiki/Antenna_array

Estimation of link budget calculation for vivaldi antenna array

$$P_R = P_T G_T G_R \lambda^2 / (4\pi R)^2$$

P_R Received power
P_T Transmitted power
G_T Transmitter antenna gain
R range
λ wavelength
G_R Receive antenna gain

- Hopely recieve signals from 10-15 meters away

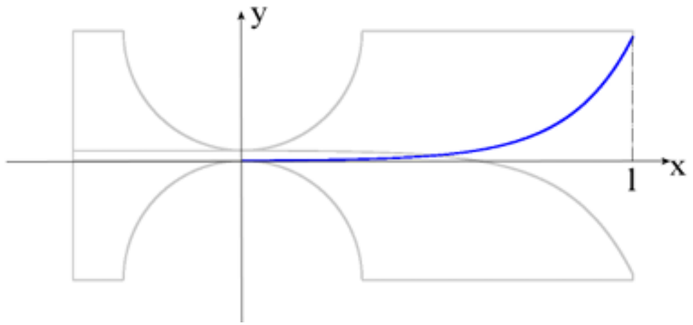
Free space path loss for 2.4 GHz frequency signal

FSPL = 20 log (4π(15)/0.125)
= **63.58 dB**

$P_R = P_T + G_T + G_R - FSPL$ (In dBm)
P_R = 6dB - 63.58dB

https://www.analog.com/en/resources/technical-articles/radio-linkbudget-calculations-for-ismrf-products.html?

Antipodal Vivaldi design



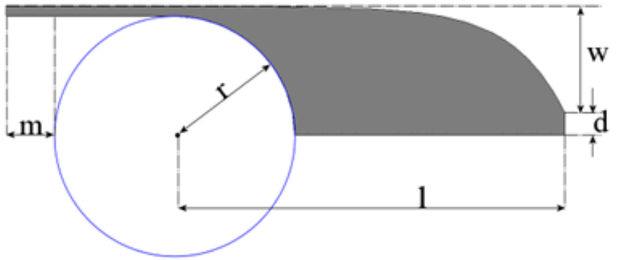


Fig. 2. Top metal layer in the Antipodal Vivaldi Antenna.

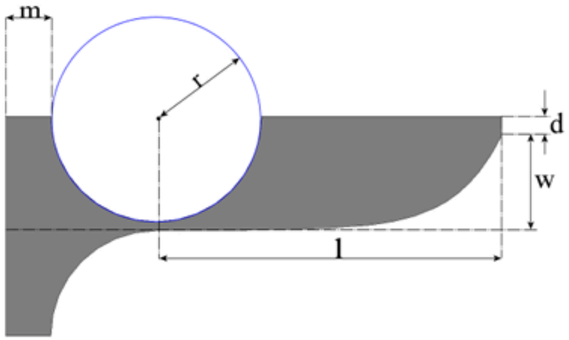
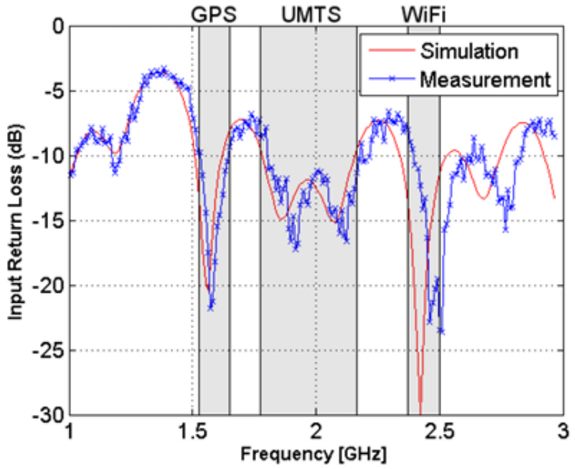


Fig. 3. Lower metal layer in the Antipodal Vivaldi Antenna.



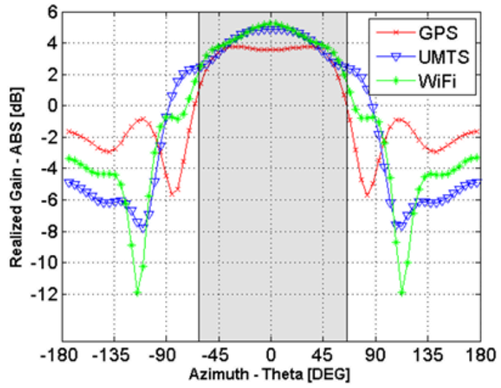


Fig. 7. Realized Gain vs. Azimuth.

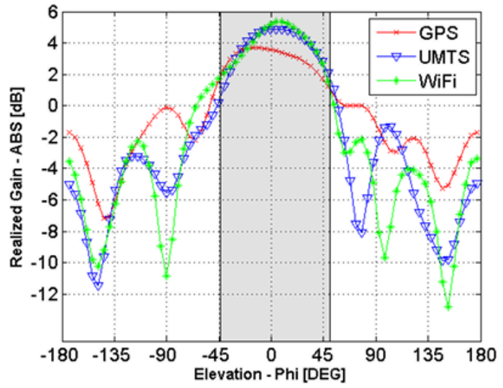
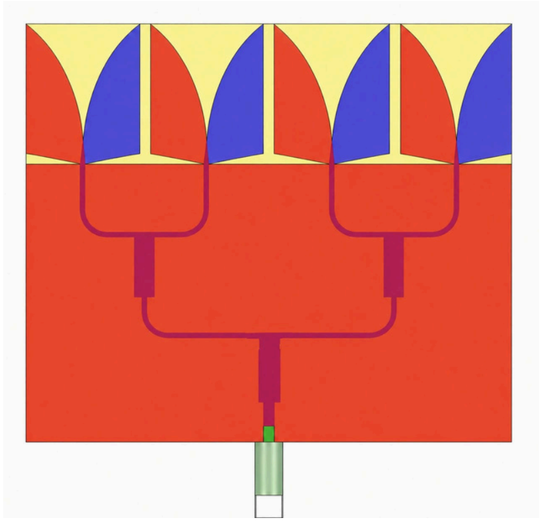


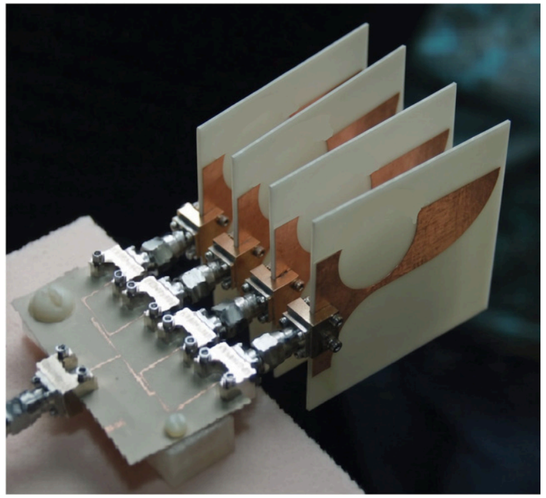
Fig. 8. Realized Gain vs. Elevation.

G. Vinci and R. Weigel, "Multiband Planar Vivaldi Antenna for Mobile Communication and Industrial Applications," 2010 International Conference on Electromagnetics in Advanced Applications, Sydney, NSW, Australia, 2010, pp. 93-96

Antipodal Vivaldi design



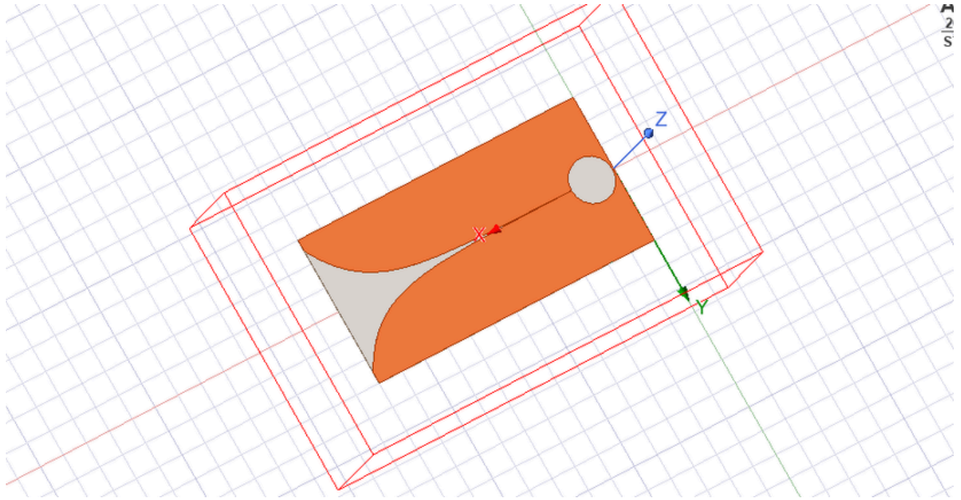
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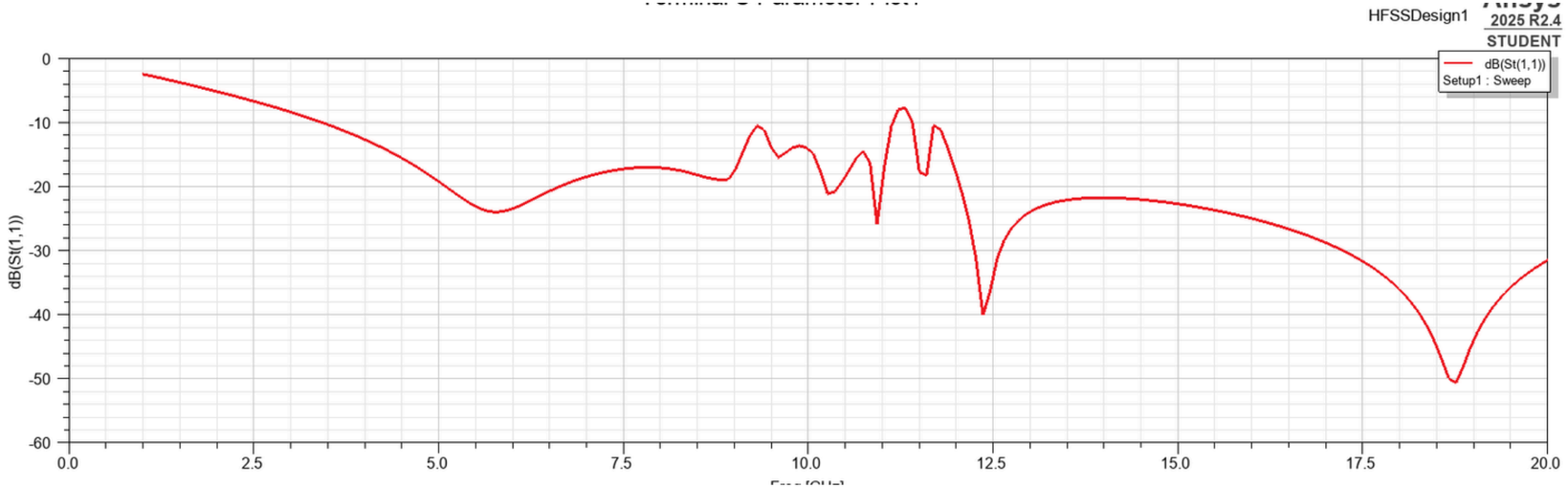
2

The second orientation is preferred because it allows the inter-element spacing d to be maintained close to the desired λ/2 spacing. In the first orientation, the physical dimensions of the Vivaldi elements constrain the placement, resulting in d>λ/2.

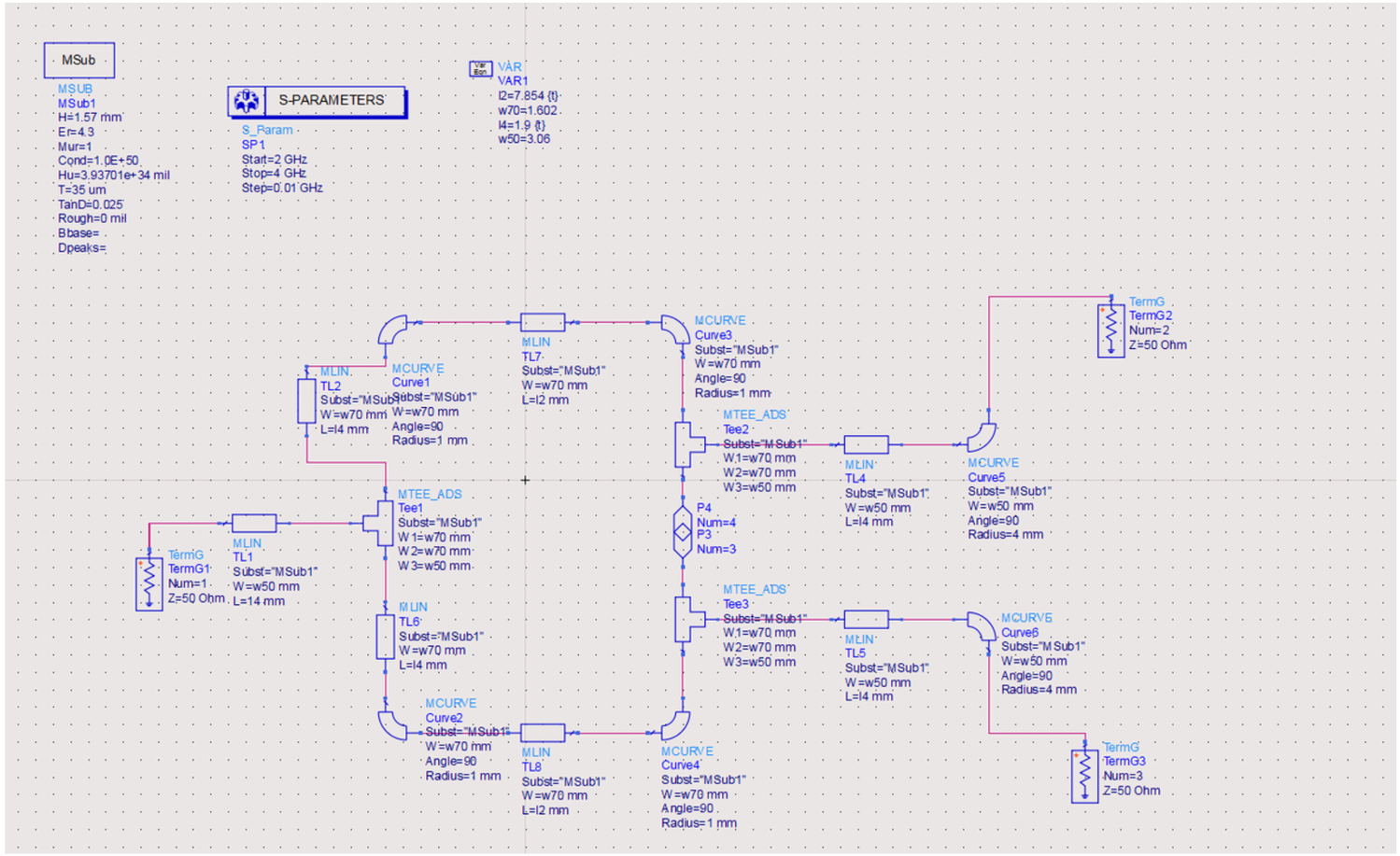
Tappered slot Vivaldi antenna simulation

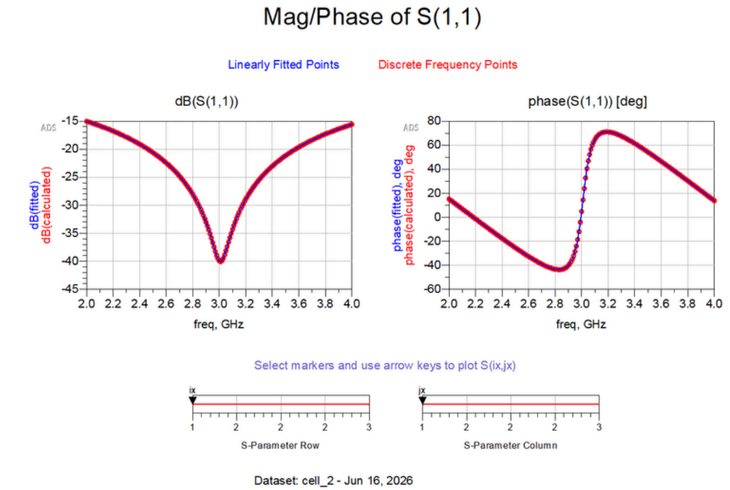


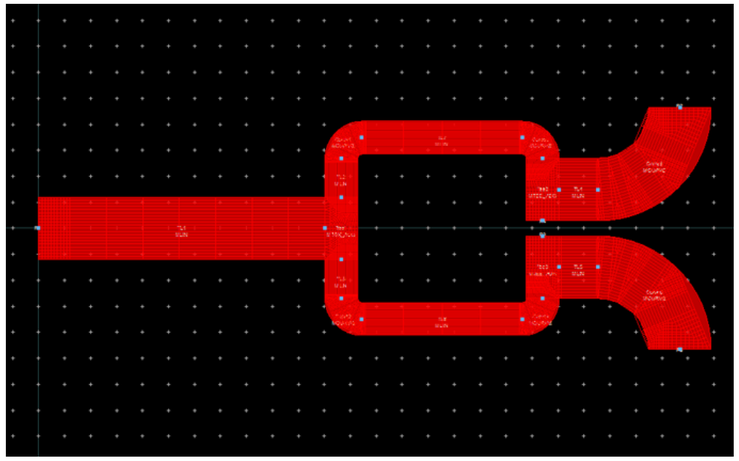
Name	Value	Unit	Evaluated V...	Type
s	0.015	mm	0.015mm	Design
rate	33.85		33.85	Design
h	0.508	mm	0.508mm	Design
TL	200	mm	200mm	Design
cThick	0.035	mm	0.035mm	Design
w	120		120	Design
a	120	mm	120mm	Design
L	240	mm	240mm	Design
MW	1.78	mm	1.78mm	Design



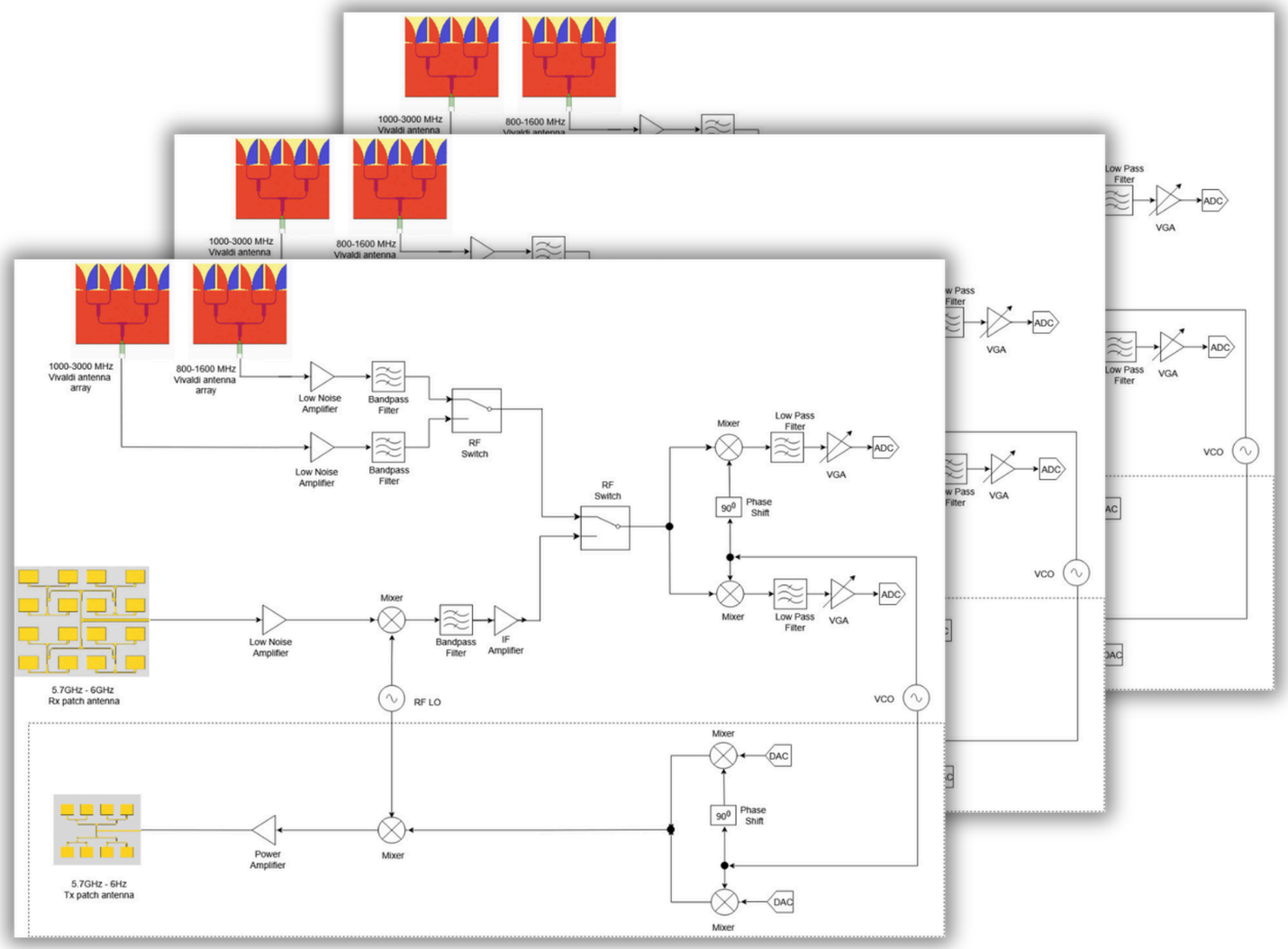
Wilkinson power divider simulation







Reciever architecture for beamforming (RF front end)



How to use OTFS for drone detection ? - Basic analogy.